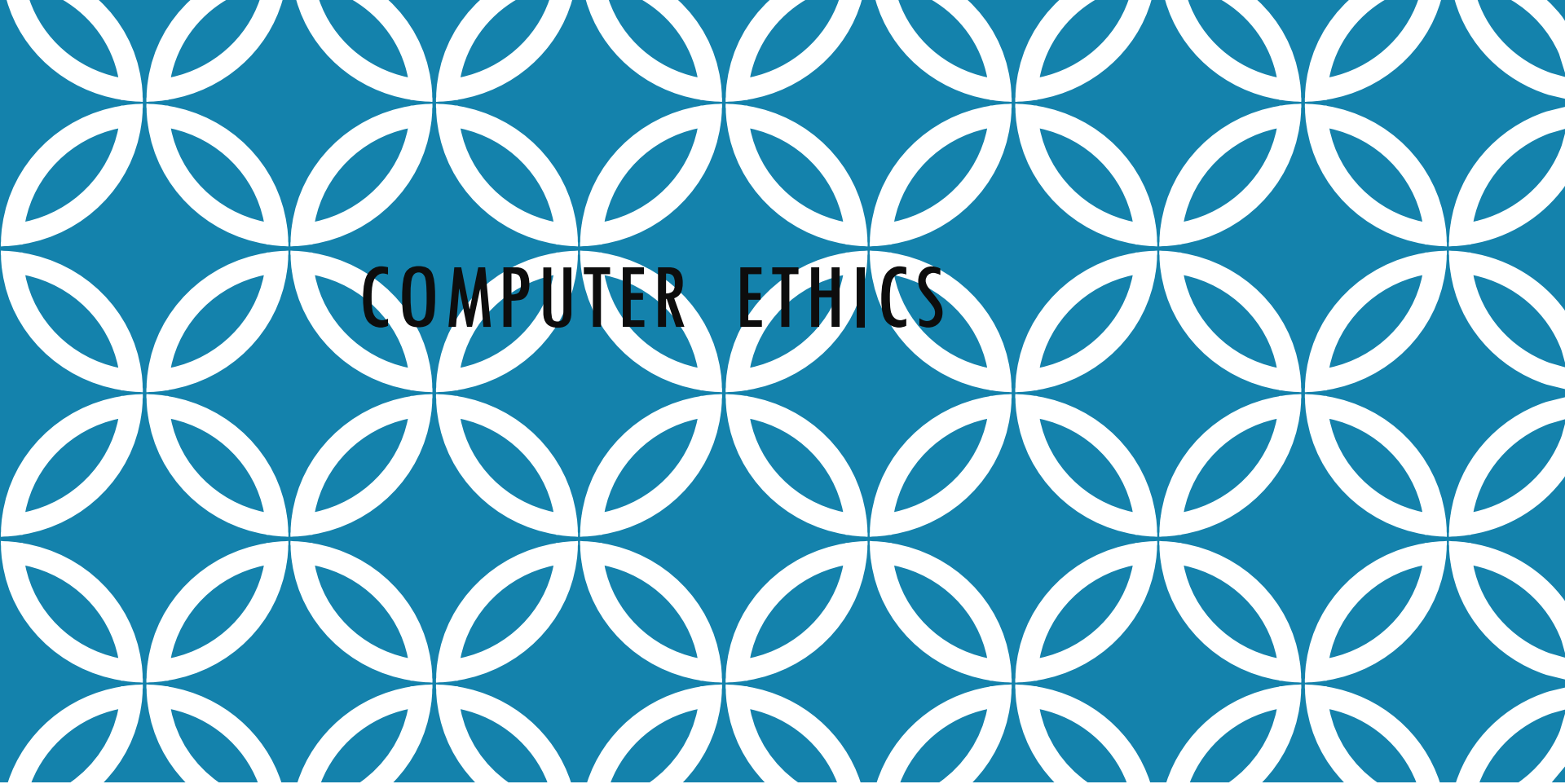




COMPUTER ETHICS

Chapter 5

Computer Abuse

Prof. Alaa El-Halees



OUTLINE

- Introduction
- Computer Fraud
- Phishing
- Spam
- Defamation
- Malware
- Hackers
- Social Engineering

INTRODUCTION



- The word **abuse** means that something is used beyond its normal use.
- Computer abuse refers to the use of computer that is seen as **harmful**.
- Computer abuse is the use of a computer to do something **improper** or **illegal**.

INTRODUCTION

- Computer abuse refers to any improper use of a computer or its resources, regardless of whether it is illegal or not.
- Computer abuse can have negative consequences, such as decreased productivity or reduced system performance, but it is not necessarily a criminal act.

INTRODUCTION

For example:

The user may be using the computer to shop for personal items during company time despite knowing that this is against the established acceptable usage policy.

COMPUTER CRIME AND COMPUTER ABUSE

Computer crime or also called Cybercrim—

Illegal acts in which computers are the primary tool.

Examples:

- Identity fraud (where personal information is stolen and used).
- Theft of financial or card payment data.
- Theft and sale of corporate data (Trade secrets).
- Ransomware attacks (demanding money to prevent a threatened attack).

COMPUTER CRIME AND COMPUTER ABUSE

Example:

In 2014, US retailer Home Depot's point of sale systems were broken. Attackers stole 50 million personal credit cards, and for some time any credit card taken at Home Depot stores was captured and its details compromised by the attackers.

COMPUTER CRIME AND COMPUTER ABUSE

Example:

Company used some person photo in their brochure is unethical (Privacy invasion) but not illegal.

So, every computer crime is abuse, but not every abuse is crime like the example above.

AVOID COMPUTER ABUSE

To avoid computer abuse, you can take the following steps:

- 1) **Install reliable antivirus software** and keep it up-to-date.
- 2) **Use strong passwords and change them regularly.**
- 3) **Be cautious of emails and attachments** from unknown sources:
- 4) **Keep your operating system and software up-to-date.**
- 5) **Avoid sharing personal information online** (Such as your full name, home address, and phone number).
- 6) **Educate yourself:** Stay informed about the latest cybersecurity threats

AVOID COMPUTER ABUSE

- 7) **Use two-factor authentication:** such as a code sent to your phone, in addition to your password. Or use USB security.
- 8) **Backup your data.**
- 9) **Be careful when downloading software:** Only download software from reputable sources. Be cautious of free software that may contain malware or spyware.
- 10) **Limit access to your computer:** If you share your computer with others, create separate user accounts for each person and limit their access to your personal files and folders.
- 11) **Monitor your accounts:** Regularly monitor your bank and credit card statements for any unauthorized transactions.
- 12) **Use encryption:** Encryption can help protect your sensitive.

AVOID COMPUTER ABUSE

12) Be cautious of social engineering tactics.

13) Use privacy settings on social media.

14) Be cautious of USB devices: USB devices can contain malware that can infect your computer

15) Use secure websites: When entering sensitive information like login credentials or credit card information online, make sure the website is secure by looking for "https" in the URL and a lock icon in the address bar.

16) Dispose of old computers properly: When disposing of old computers, make sure to securely erase all personal data and wipe the hard drive to prevent unauthorized access to your personal information.

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COMPUTER FRAUD

Computer Fraud:

Computer fraud refers to any dishonest or illegal activity that involves the use of computer technology.

COMPUTER FRAUD

For instance, imagine an employee working at a financial institution who has access to confidential information about upcoming mergers or acquisitions. This employee can use their computer skills to break security measures and access the sensitive data.

Once they have obtained the non-public information, the employee can then use it to make stock trades to their advantage.

COMPUTER FRAUD

Another example of computer fraud is account fraud.

Account fraud is a wide-ranging term or theft and fraud committed using a credit card or any bank account mechanism as a fraudulent source of funds in a transaction.

The purpose may be to obtain goods without paying, or to obtain unauthorized funds from an account.

COMPUTER FRAUD

Example

18 persons were charged in 2013 in an alleged conspiracy to create more than 7,000 fake identities to obtain "tens of thousands" of credit cards.

According to the office, they did not repay the loans, causing more than \$200 million in losses to businesses and financial institutions.

Computer Fraud

In 2018, a BBC report described how a woman lost more than £4,000 when her Royal Bank of Scotland account was taken over. In this case, the criminal didn't use online banking to access the account, but rather used a little charm and savvy (aka social engineering) to persuade one of the bank's customer service representatives to approve a fraudulent transfer of £4,318.

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PHISHING

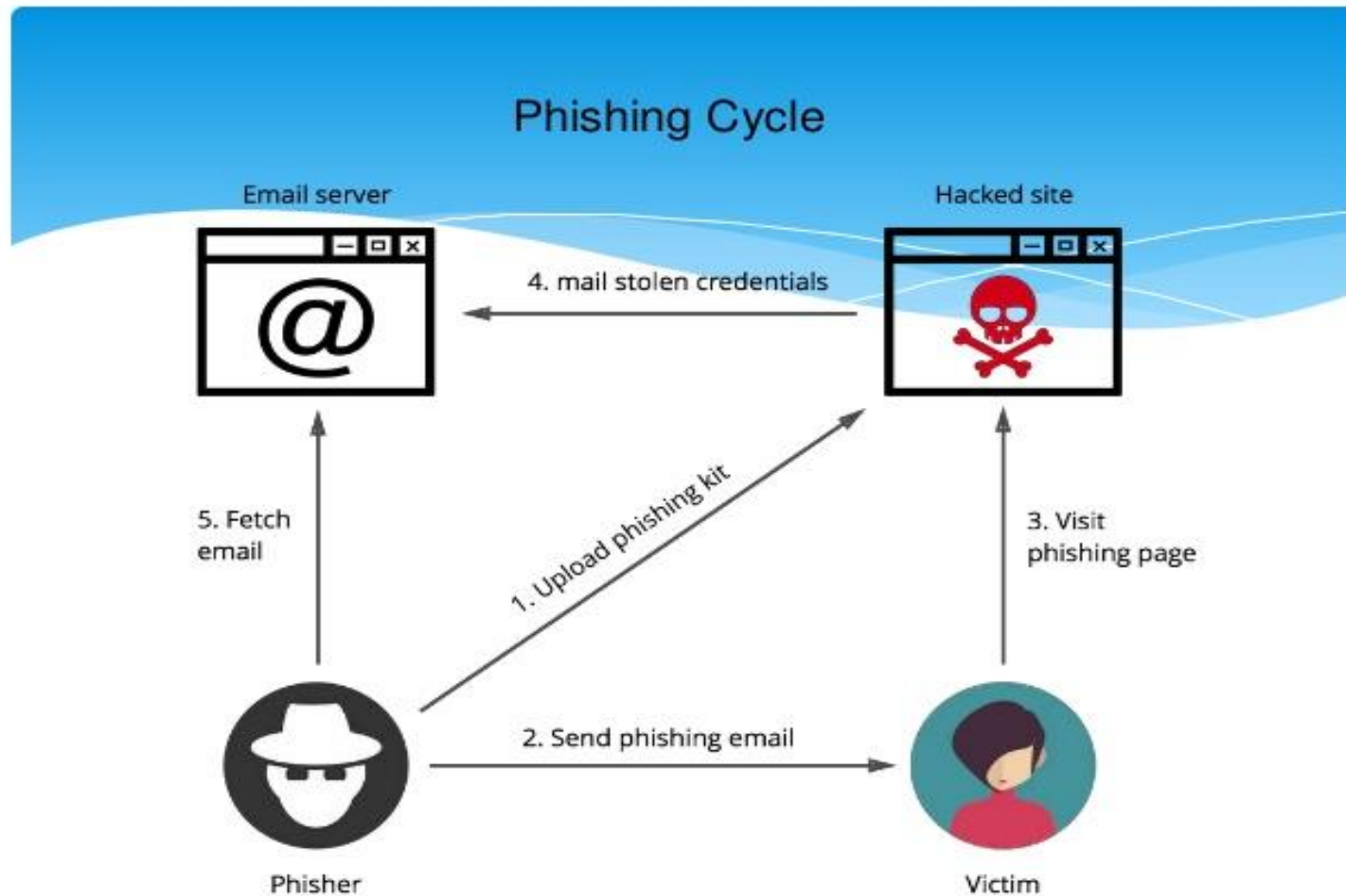
Phishing is a method of trying to gather personal information using misleading e-mails and websites.

It designed to **trick** you into giving your password or other information to an imposter.

Phishing

- For example, let's say you get an email claiming to be from **your bank**. The email says your account may be compromised, and you should click this link to take action.
- You click the link in the email and end up on a site that **looks just like your** bank's real website.
- In a rush to secure your account, you type your password and possibly other details like your credit card number. You've been phished. The attacker now has your bank account's username and password, as well as any other information you provided. That wasn't your bank's real website

Phishing



PHISHING

From: service@paypal.com
Subject: Your PayPal Account



Security Center Advisory!

We recently noticed one or more attempts to log in to your PayPal account from a foreign IP address and we have reasons to believe that your account was hijacked by a third party without your authorization. If you recently accessed your account while traveling, the unusual log in attempts may have been initiated by you.

If you are the rightful holder of the account you must **click the link below** and then complete all steps from the following page as we try to verify your identity.

[Click here to verify your account](#)

Actual link URL: [http://80.179.238.73/ ... paypal/](http://80.179.238.73/...paypal/)

If you choose to ignore our request, you leave us no choice but to temporaly suspend your account.

Thank you for using PayPal! The PayPal Team

Please do not reply to this e-mail. Mail sent to this address cannot be answered. For assistance, [log in](#) to your PayPal account and choose the "Help" link in the footer of any page.

To receive email notifications in plain text instead of HTML, update your preferences [here](#).

PayPal Email ID PP697

Protect Your Account Info

Make sure you never provide your password to fraudulent persons.

PayPal automatically encrypts your confidential information using the Secure Sockets Layer protocol (SSL) with an encryption key length of 128-bits (the highest level commercially available).

PayPal will never ask you to enter your password in an email.

For more information on protecting yourself from fraud, please review our Security Tips at <http://www.paypal.com/securitytips>

Protect Your Password

You should never give your PayPal password to anyone, including PayPal employees.

Phishing



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SPAM

Spamming is the abuse of electronic messaging systems to send *unsolicited bulk* messages.

Unsolicited means that the Recipient has not granted verifiable permission for the message to be sent.

Bulk means that the message is sent as part of a larger collection of messages, all having identical content .

SPAM

Email Spam

Email spam usually pertains to unsolicited commercial messages sent in bulk by people you don't know

Spam is mostly used to advertise products and sometimes to broadcast some political or social commentary.

Spam can be considered as the most annoying **cyber-pollution** that targets all of us with tons of unsolicited emails.

Spam

Dealing with spam become one of the Internet's biggest problem.

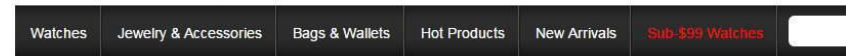
In 2021, it was estimated that nearly 319.6 billion emails were sent and received daily.

spam makes up 45% of all emails.

Spammers receive 1 reply for every 12,500,000 emails sent.

SPAM

Buy Now!
\$14.95
One Year **FREE** Upgrades



SHOP BY BRAND

- >Rolex Sports
- >Rolex Dress
- >Louis Vuitton
- >Breitling
- >Bvlgari
- >Cartier
- >Chopard
- >Dior
- >Ebel
- >Enicar
- >Ferrari
- >Franck Muller

FEATURED PRODUCTS

Rolex Sports

18Kt/SS Two Tone Rolesium Yachtmaster



\$483.00 > more details

Rolex Dress

Ladies' Full Stainless Steel Jubilee Datejust



\$434.00 > more details

Rolex Sports

18Kt/SS Two Tone GMT Master II



\$493.00 > more details

SPAM

Types of Spam:

While the most widely recognized form of spam is e-mail spam, the term is applied to similar abuses in other media: Social network spam, chat spam, Web search engine spam, spam in blogs, wiki spam, mobile phone messaging spam and Internet forum spam.

SOCIAL NETWORKING SPAM



Policy Categories: Spam



Jay Brown

Comment at 12/12/15 4:57:15 AM

I help people who are interested in working from home.. you can make \$500 - \$2000 per week all you need is internet, if you don't have a computer you can also use your phone, or tablet.. We only have 20 positions available so add me and message me if you are interested. PS this is for the United States and Canada ONLY! add me then message me.

[See less](#)

[View on Facebook](#) | Posted via: Web

SPAM BOT

- A spam bot is a type of computer program designed to automatically send spam messages, typically through email or social media, to a large number of recipients.
- These messages may contain advertisements, phishing links, malware, or other unwanted content.

SPAM BOT



I love this recipe! My trick is to add lemon at the end :)



Is there a gluten free version of this dish?



Are YOU ready to Make BIG \$\$\$\$ by working from HOME!!?? Click this link to start making BANK from ur COUCH: www.dangerous-site.com



I use rice flour instead of regular to make it gluten free!

SPAM AND FLAME

Flaming vs. Spamming

Spamming is causing one or more email to be flooded with irrelevant or inappropriate messages.

Flaming is to send hate or insulting messages via e-mail or in other forms of online communication such as chat rooms.

SPAM AND FLAME

Example of flame:

Let's say someone posts a video on social media about their favorite hobby, and someone else comments:

"Wow, you really have no life. Who cares about your stupid hobby?"

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DEFAMATION



Defamation

The intention to harm the reputation of a particular person or reasonably know that his/her behavior could cause such harm.

Courts have to balance the reputation of one person with the free of expression rights of the other.

Example if someone writing that another person is a thief, if the written statement is not true then the writer can be suited as defendant.

DEFAMATION

Defamation in information technology refers to the publication or distribution of false and damaging statements about an individual or organization through electronic communication channels such as the internet, social media, or email.

DEFAMATION

Example: James Woods, an American actor, sued a number of Twitter users for defamation after they accused him of being a "cocaine addict" on the social media platform. Woods denied the allegations and claimed that the users had intentionally spread false and defamatory information about him.

DEFAMATION

Elements of Defamation

- Defendant's false statement of fact about person or organization ...etc.
- Publication of this statement to a **third party**.
- This publication causes damage to the person.

DEFAMATION

Why defamation in the Internet is more damaging?

Because rumors move at light speed to huge audiences and can cause a great deal more damage than what happens in other ways.

Why users of the Internet are more likely than ordinary citizens to be found publishing comments which are actionable as defamatory?

Because on the Internet it is very easy to be anonymous. If people don't know who you really are, then it is very difficult for anyone to find you.

DEFAMATION

Example:

In 2015 North Carolina defamation case, Hammond posted in a Facebook group implying her former business associate, Dial, *had gotten drunk and caused the death of her own child.*

Dial's child had actually been killed 39 years prior in a tragic accident that had nothing to do with Dial. This defamatory comment about Dial accused her of the **felony crime of manslaughter**. Due to the emotional distress she suffered from these claims, she sued Hammond for defamation. *She won an unusually high \$500,000 settlement*

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MALWARE

Malware is software designed to damage a computer system without the owner's informed permission. The term comes from the words **malicious** and **software**.

It is any software intentionally designed to cause damage to a computer, server, client, or computer network:

MALWARE

Some common purposes of malware include:

1- Stealing sensitive information: Malware can be designed to steal personal or financial information, such as login credentials, credit card numbers, or social security numbers.

2- Disrupting computer systems: Malware can be used to disrupt computer systems or networks, causing them to crash or become unavailable.

3- Gaining unauthorized access: Malware can be used to gain unauthorized access to a system or network, allowing the attacker to control or manipulate the system or steal sensitive data.

MALWARE

5. Generating revenue: Malware can be used to generate revenue for the attacker through means such as ransomware, adware, or cryptocurrency mining.

6.Espionage: Malware can be used for espionage purposes, such as stealing intellectual property or confidential business information.

MALWARE

A wide variety of types of malware exist, including

Virus

Worm

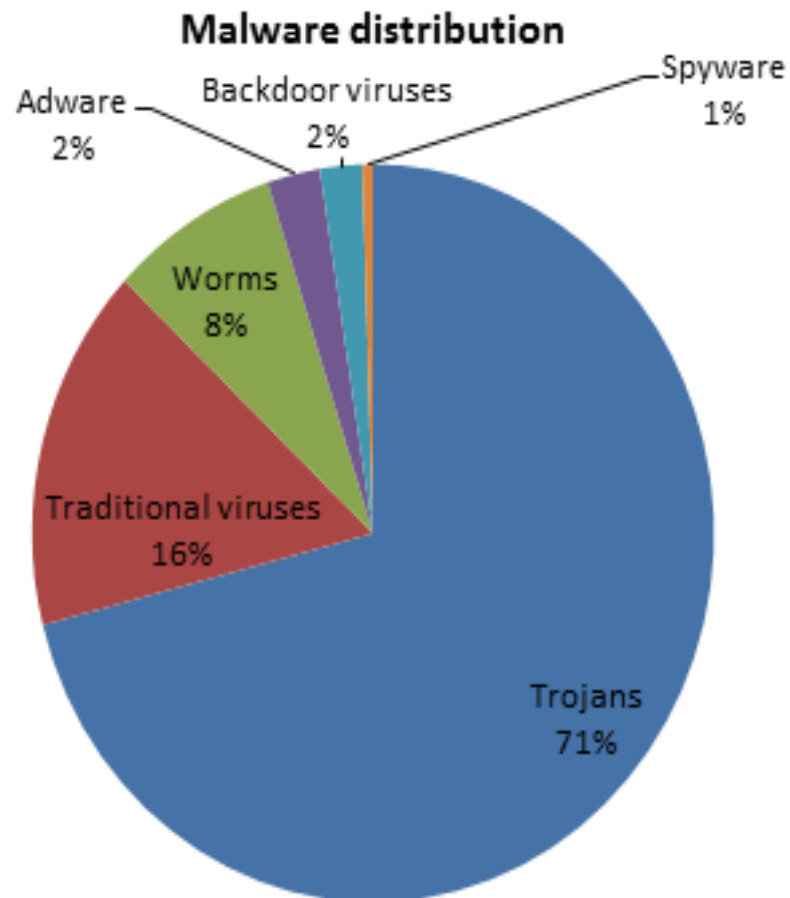
Trojan horse

Adware

Spyware

Botnets

MALWARE



VIRUS

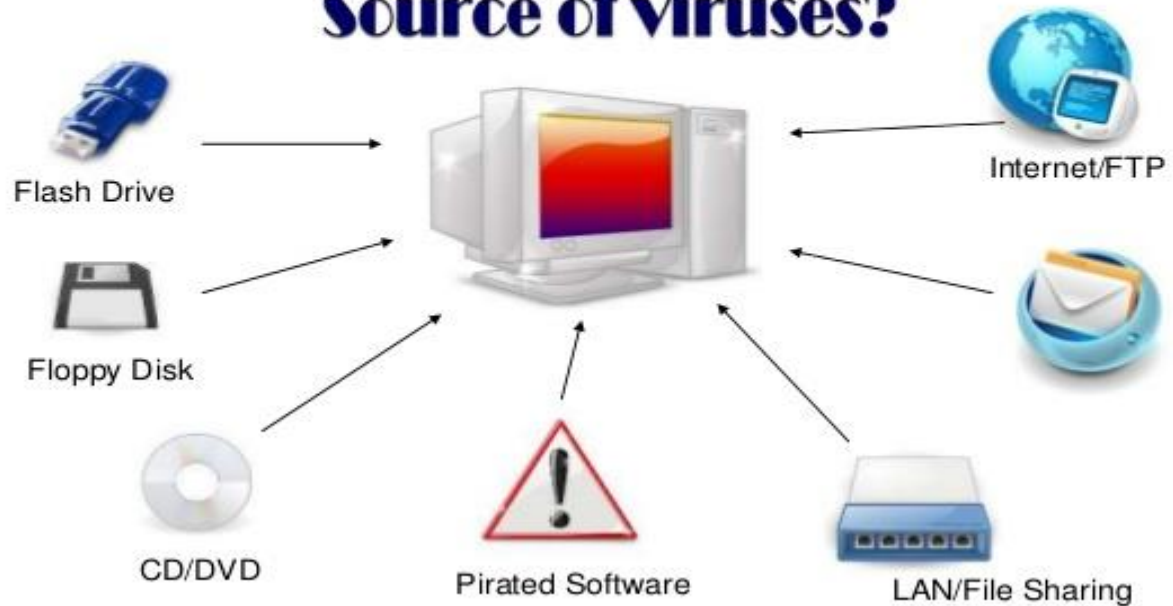
1) Virus:

- Computer viruses are small code that are designed to spread from one computer to another and to interfere with computer operation.
- Computer viruses typically attach to an executable host file, which results in their viral codes executing when a file is opened .
- A virus might corrupt or delete data on your computer, or even erase everything on your hard disk

VIRUS

Sources of virus?

Source of viruses?

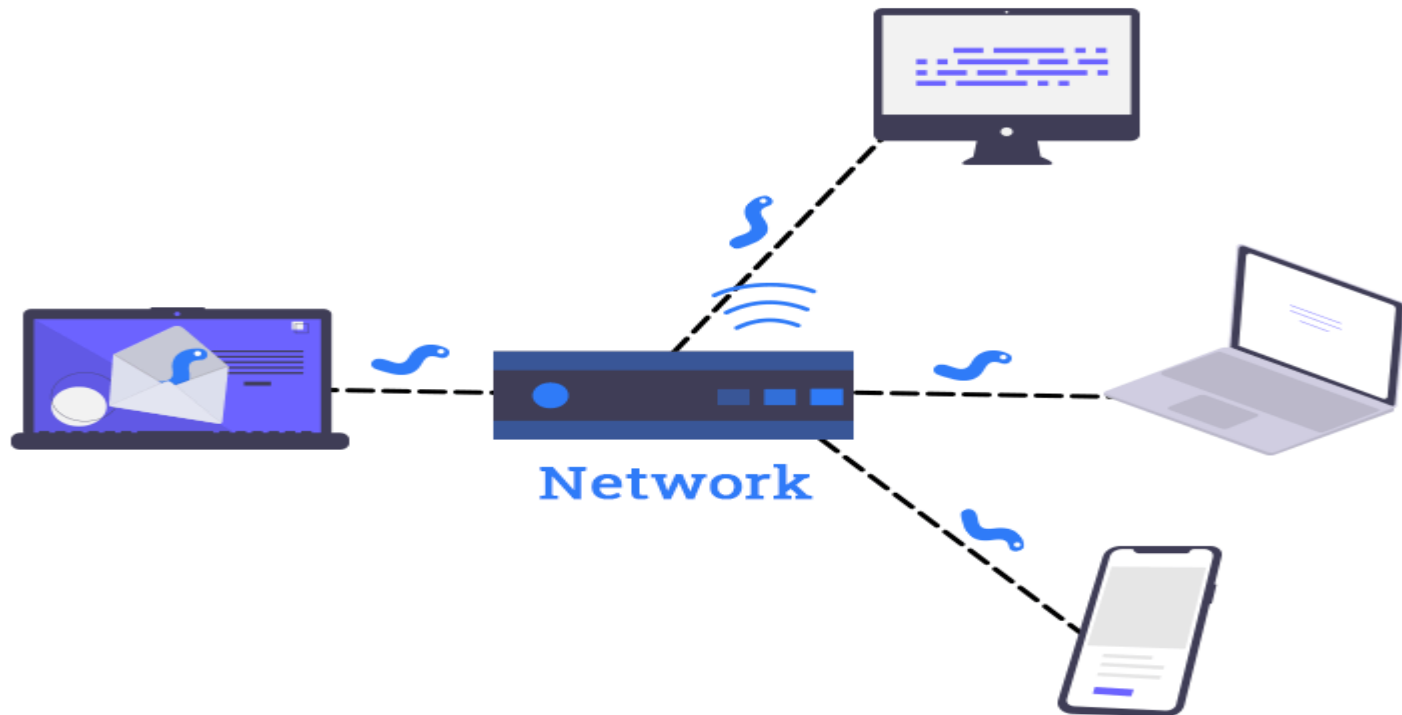


WORMS

2) Worms:

- Self-contained program
- Spreads through a computer network
- Exploits security holes in networked computers
- **worm** is similar to a virus by its design, and is considered to be a sub-class of a virus. Worms spread from computer to computer, but unlike a virus, it has the capability to travel without any help from a person.

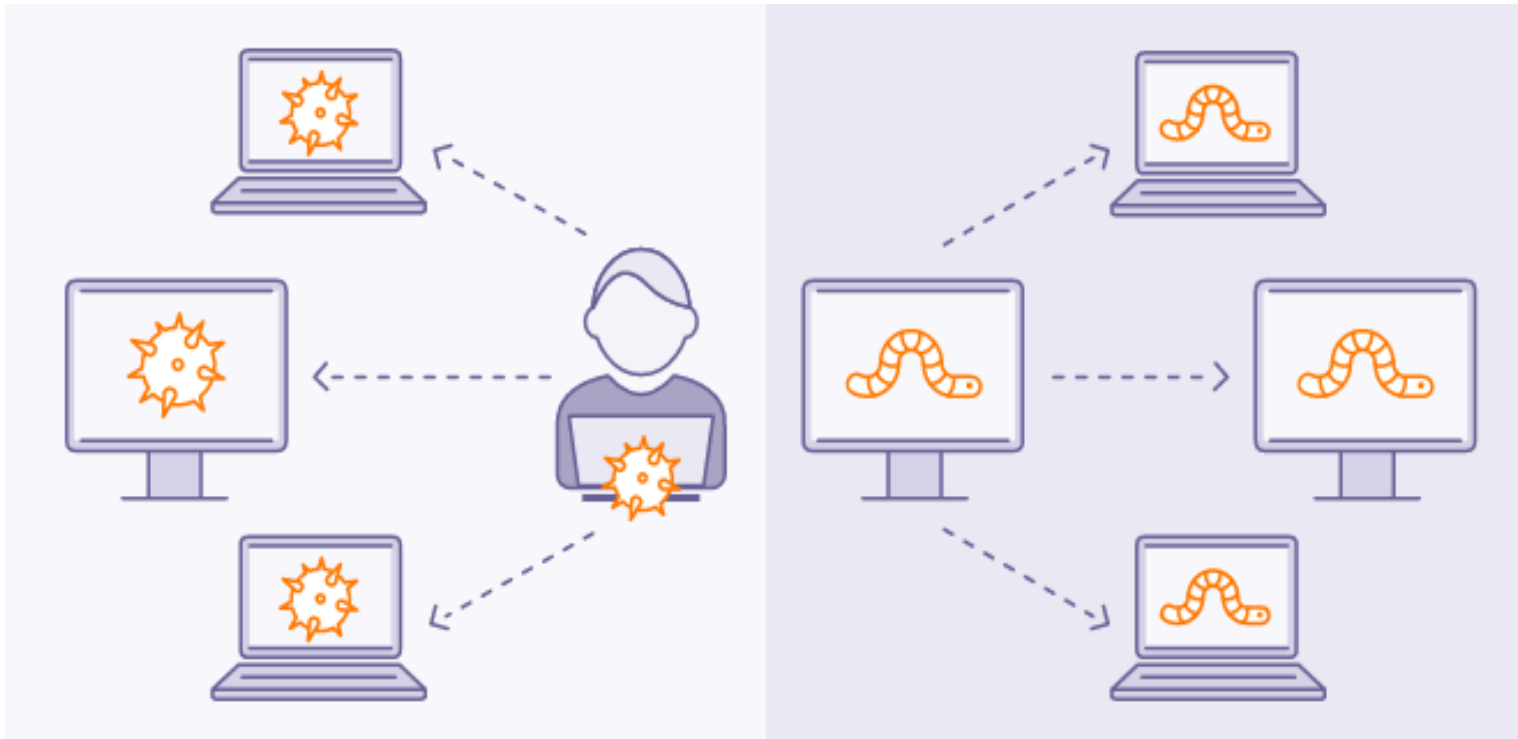
WORMS



WORMS

One example would be for a worm to send a copy of itself to everyone listed in your e-mail address book. Then, the worm replicates and sends itself out to everyone listed in each of the receiver's address book, and the manifest continues on down the line

WORM VS. VIRUS



TROJAN HORSES

3) Trojan horses

A Trojan horse is a type of malicious software that is designed to look like a legitimate program or file, but actually contains hidden malicious code.

Trojan horses are **non-replicating** programs that perform some unwanted action while **pretending to be useful**.

Trojan horses activate when they are executed.

TROJAN HORSES

The basic difference from computer viruses is: a Trojan horse is technically a normal computer program and does not possess the means to spread itself. Originally Trojan horses were not designed to spread themselves. They relied on fooling people to allow the program to perform actions that they would not have voluntarily performed

TROJAN HORSES

Example of Trojan Horse

Fake Antivirus Trojans: These Trojans pose as legitimate antivirus software, tricking users into downloading and installing them. Once installed, they can display fake virus alerts and demand payment for their removal, but they do not actually remove any real viruses.

TROJAN HORSES

In 2011, the computers in the Japanese parliament building were infected with a Trojan horse allegedly created by the Chinese government. The Trojan was installed after a member of the parliament opened an infected email

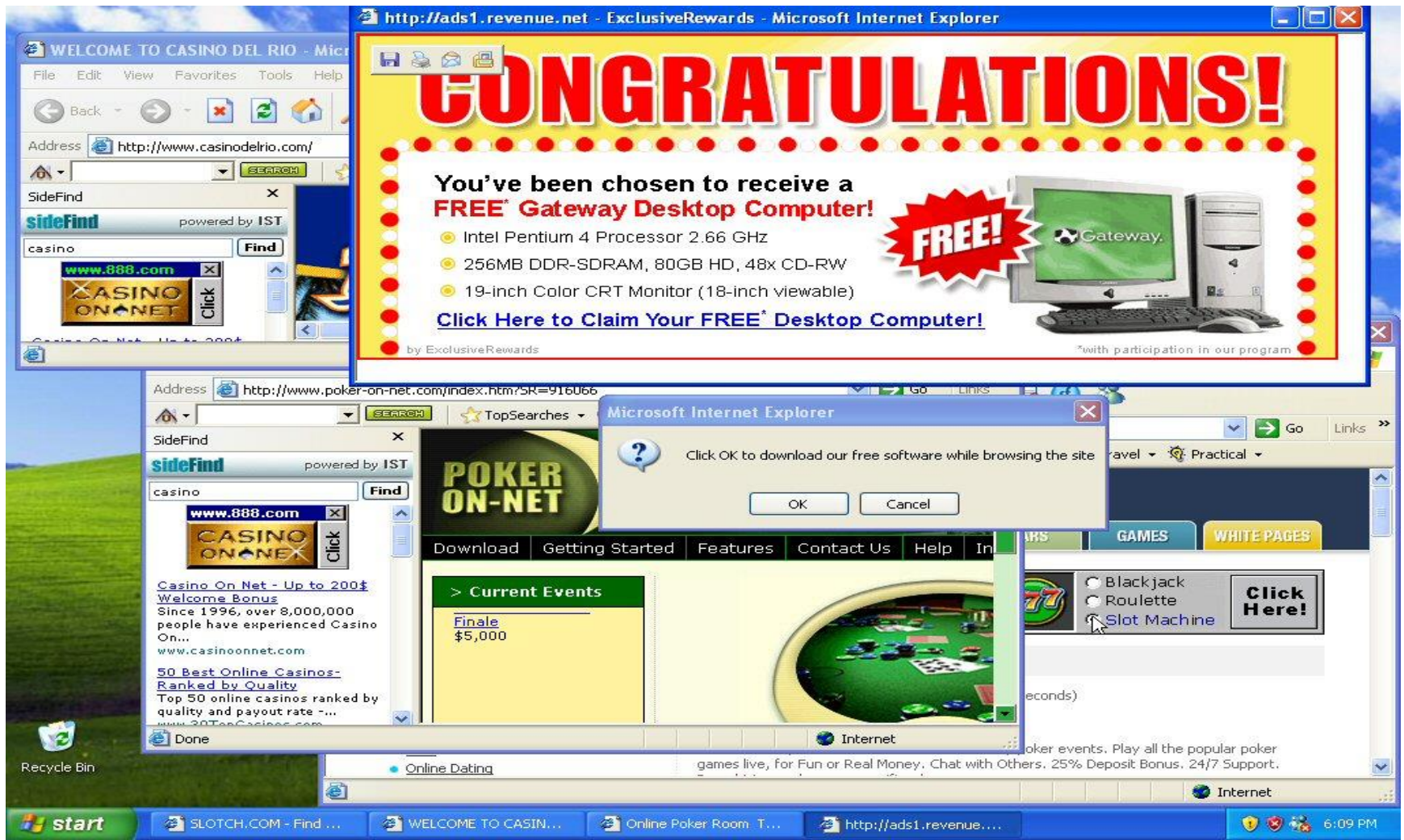
MALWARE

4) Adware

Adware or **advertising-supported software** is any software package which automatically plays, displays, or downloads advertising material to a computer after the software is installed on it or while the application is being used.

The authors of these applications include **additional code** that delivers the ads, which can be viewed through pop-up windows or through a bar that appears on a computer screen.

ADWARE



ADWARE

There are two main ways by which adware enter onto your system.

first one, by downloading a program—usually **freeware** or **shareware**—and it quietly installs adware without your knowledge, or permission.

second method is by visiting a website. It can be infected with adware, which takes advantage of a vulnerability in the user's web browser to deliver a **drive-by download**.

MALWARE



5) Spyware:

Spyware is a type of malicious software designed to secretly collect information about a computer user's activities without their knowledge or consent.

SPYWARE

Spyware can monitor a user's online behavior, including the websites they visit, the searches they make, and the information they enter into online forms. It can also capture personal information, such as passwords, credit card numbers, and other sensitive data.

SPYWARE

Spyware is often pushed with free software downloads, email attachments, or malicious websites. Once installed, it can be difficult to detect and remove, as it often operates in the background without the user's knowledge.

SPYWARE

Example:

PZChao is a type of spyware that was discovered in 2017. It was spread through malicious email attachments and targeted organizations in the financial, telecommunications, and government sectors. Once installed on a victim's computer, PZChao was able to steal sensitive data, log keystrokes, and take screenshots.

BOTNETS

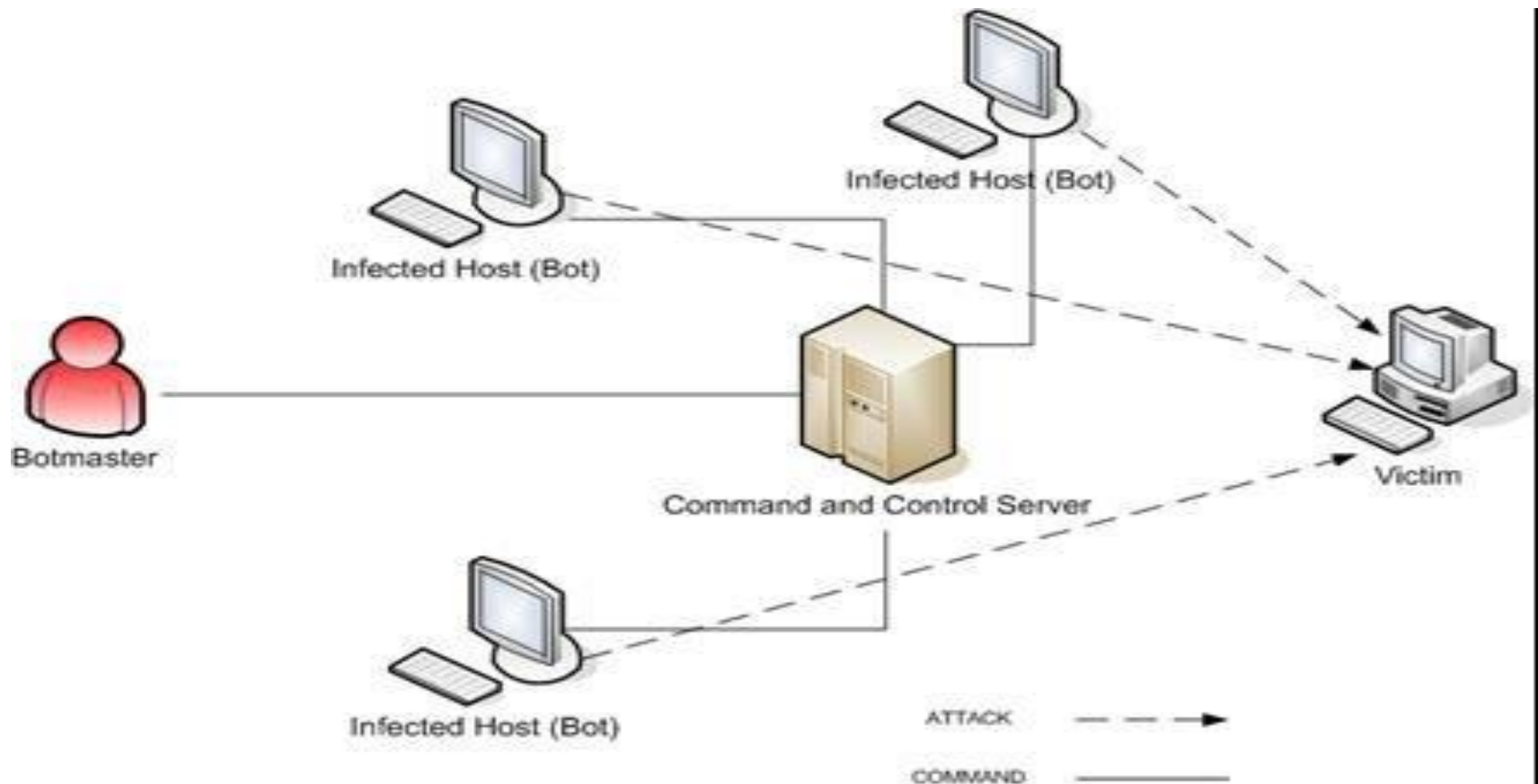
6) Botnets: Malware can be used to create a network of infected computers known as a botnet.

Botnets can be used for various purposes, such as launching DDoS attacks, sending spam emails, or mining cryptocurrency.

BOTNETS

Once a device is infected with a bot, the malware allows the device to be remotely controlled by the botnet's operator, who can then use the device to perform a variety of tasks without the owner's knowledge or consent.

BOTNET



BOTNETS

Example:

Mirai: In 2016, the Mirai botnet infected hundreds of thousands of Internet of Things (IoT) devices, including routers, security cameras, and DVRs. The botnet was used to launch massive DDoS attacks against DNS provider Dyn, which caused widespread internet disruption across the United States.

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HACKING

- **Hacking** is a term used to describe the act of using computers, networks, or other technological systems to gain unauthorized access to data, systems, or resources.

HACKING

A computer hacker uses his or her skills to gain unauthorized access to another computer or computer network and, once inside, can alter important information, deleting essential files, or just crashing the whole system

Many hackers who break into commercial Web sites are looking for customers' private financial data (including bank account and credit card numbers)

Stealing money by hacking consider as computer Crime.

HACKERS

Types of Hackers:

Nowadays there are many type of hackers:

- 1) **White hackers** (also known as ethical hackers) are those who use their knowledge for selling their services to clients who want to protect their networks.
- 2) **Black hackers** (also known as crackers) are the ones who attack those networks and try to make some money out of it.

HACKERS

3) **Grey hackers** They don't do it for the money; they do it for the show.

They want to show the world and their hacker friends how good they are.

4) **Industrial spy:** Hacker to Capture trade secrets and gain competitive advantage.

5) **Hacktivist:** is hacking to achieve a political or social goal.

6) **Cyberterrorist** Destroy infrastructure components of financial institutions, utilities, and emergency response units

HACKERS

Hacktivist Example:

"Palestinian Electronic Army" (PEA), which is a group of hackers who use their technical skills to promote the Palestinian cause and expose the actions of the Israeli government.

The PEA has been involved in several attacks against Israeli targets, including government websites, social media accounts, and other online platforms.

The group using tactics such as defacing websites, leaking sensitive information, and launching DDoS attacks.

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SOCIAL ENGINEERING

Social engineering relies on human nature, rather than technical hacking, to manipulate people into compromising personal or enterprise security.

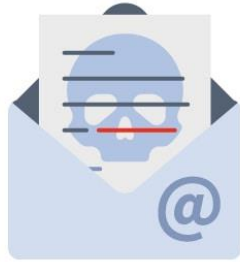
SOCIAL ENGINEERING

Attackers may use psychological tactics to make their targets more likely to comply with their requests, such as creating a sense of urgency or authority.

For example, instead of trying to find a software weakness to hack a system, a social engineer might call an employee and pretend to be as an IT support person (for example) , trying to trick the employee into telling his password.

Social Engineering Tactics to Watch For

Knowing the red flags can help you avoid becoming a victim.



Your 'friend' sends you a strange message.



Your emotions are heightened.



The request is urgent.



The offer feels too good to be true.



You're receiving help you didn't ask for.



The sender can't prove their identity.

SOCIAL ENGINEERING

Computers, networks and software applications have been **hardened** – The human being has become the weak link in security

SOCIAL ENGINEERING

There are many types of social engineering attacks,

For example:

- Phishing (we talked about earlier)
- Pretexting
- Baiting
- Scareware

SOCIAL ENGINEERING

2) **Pretexting**: uses false identity to trick victims into giving up information.

Pretexting is a form of social engineering where attackers focus on creating a good pretext, or a fabricated scenario, that they use to try and steal their victims' personal information

SOCIAL ENGINEERING

Pretexters can pretend to be co-workers, police officers, bankers, tax authorities, insurance investigators, etc. Simply put: anyone who can be perceived as having authority or a right-to-know by the targeted victim.

The pretexter must simply prepare answers to questions that might be asked by the victim. Sometimes, an authoritative voice, a serious tone, and an ability to think on one's feet are all that is needed to create a *pretextual scenario*.

SOCIAL ENGINEERING

A scammer recently used a mix of pretexting scenarios and an AI-generated voice of the CEO of a German firm to convince the CEO of the UK subsidiary to transfer a sum of \$243,000 to a Hungarian supplier. The UK subsidiary CEO thought he had gotten a call from the CEO of the parent company in Germany. According to the Wall Street Journal report, the AI-generated call accurately mimicked the voice and German accent of the impersonated CEO, enough to get the UK subsidiary CEO to recognize it as his boss's voice.

SOCIAL ENGINEERING

3) Baiting: Baiting is when an attacker leaves a malware-infected physical device, such as a USB flash drive, in a place it is sure to be found. The finder then picks up the device and loads it onto his or her computer, unintentionally installing the malware.

Social Engineering

Example:

- Steve , VP and founder of Secure Network Technologies, was hired by a credit union to analyze security.
- He decided to load a Trojan onto a few flash drives and place them in the parking lot.
- Within a few hours the Trojan was found on several computers and was transmitting data from the **credit union computers** to a remote anonymous email account.

Social Engineering

4) Scareware

As the name indicates, **scareware** is malware that's meant to scare you to take action — and take action fast.

It often comes in the form of pop-ups or emails indicating you need to “act now” to get rid of viruses or malware on your device. In fact, if you act you might be downloading a **computer virus** or malware.

Social Engineering

Example

In 2017, a scareware campaign called "Mac Defender" specifically targeted Mac users. Users would encounter fake pop-up advertisements while browsing the internet, stating that their Mac computer was infected with viruses or malware.

Social Engineering

The pop-ups would mimic legitimate macOS system alerts, alarming users with messages about potential data loss or system damage. To resolve the issue, users were prompted to download and install the supposed antivirus software called "Mac Defender."

However, instead of a genuine antivirus program, the software was malicious in nature. Once installed, it would display fake infection reports and continuously prompt users to enter their credit card information to purchase the full version of the software for threat removal.

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<https://www.impactlaw.com/criminal-law/white-collar/computer-internet-fraud>